

AI-Based Enterprise Knowledge Assistant

PHASE I REPORT

Submitted by

JANANI KUMARESH	230701122
M. NITHYASHREE	230701221
S. NITHYASHREE	230701220

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

ANNA UNIVERSITY: CHENNAI 600 025

December 2026

ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Report titled “**AI-BASED ENTERPRISE KNOWLEDGE ASSISTANT**” is the bonafide work of **JANANI KUMARESH (230701122)**, **M.NITHYASHREE (230701221)**, **S.NITHYASHREE (230701220)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidates.

“This project addresses the following Sustainable Development Goal(s): SDG 4 – Quality Education”

SIGNATURE

Dr. J. Manoranjini,

Professor and Head,
Department of Computer Science
and Engineering,
Rajalakshmi Engineering College,
Chennai – 602 105.

SIGNATURE

A. Poornima,

Assistant Professor
Department of Computer Science
and Engineering,
Rajalakshmi Engineering College,
Chennai – 602 105.

Submitted to Project Viva-Voce Examination held on _____

Internal Examiner

External Examiner

ABSTRACT

Accessing accurate administrative and faculty details in educational institutions is a frustrating and time-consuming exercise. As department locations, room numbers, faculty office addresses and contact details are distributed across physical notice boards, website pages and PDF circulars, students, particularly new ones, waste extensive time searching on campus for these details. This work presents an Enterprise Knowledge Assistant that consolidates this information into a single interface and allows users to search these details in natural everyday language and receive accurate and targeted answers. The system is built with institutional data structured as high-dimensional vector embeddings and stored in a vector database to allow efficient retrieval, aided by Retrieval-Augmented Generation (RAG) and semantic search to ground the answers in verified institutional data. As an instant knowledge repository, the assistant reduces the search time of students and reduces repetitive queries to administrative staff, improving daily campus operations. This work showcases how domain-specific Retrieval-Augmented Generation systems can aid institutional knowledge access and enhance the student experience.

Keywords: Enterprise Knowledge Assistant, Retrieval-Augmented Generation (RAG), Semantic Search, Natural Language Processing (NLP), Vector Embeddings, Large Language Models (LLMs).

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.,** our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.,** and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.,** for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.,** our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. Manoranjini, Ph.D.,** Professor and Head of the Department of Computer Science and Engineering for her guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide,

A. Poornima, Assistant Professor, Department of Computer Science and Engineering for her valuable guidance throughout the course of the project. We thank our Project Coordinator, **Dr.S.Madhusudhanan.,** Associate Professor, Department of Computer Science and Engineering for his useful tips during our review to build our project.

JANANI KUMARESH : 230701122

M. NITHYASHREE : 230701221

S. NITHYASHREE : 230701220

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
	LIST OF TABLES	vii
	LIST OF FIGURES	viii
	LIST OF ABBREVIATIONS	ix
1.	INTRODUCTION	1
	1.1 GENERAL	1
	1.2 OBJECTIVE	2
	1.3 EXISTING SYSTEM	3
	1.4 PROPOSED SYSTEM	4
2.	LITERATURE SURVEY	5
	2.1 LITERATURE REVIEW	5
	2.2 COMPARISON AND DISCUSSION	10
	2.3 CONCLUSION	11
3.	SYSTEM DESIGN	12
	3.1 SYSTEM ARCHITECTURE	12
	3.2 DATA INGESTION AND INDEXING	14
	3.3 SEMANTIC RETRIEVAL AND RESPONSE PIPELINE	14

4.	PROJECT DESCRIPTION	15
	4.1 METHODOLOGIES	15
	4.2 MODULES	20
	4.2.1 DATA INGESTION MODULE	20
	4.2.2 RAG RESPONSE GENERATION MODULE	21
5.	IMPLEMENTATION AND RESULT DISCUSSION	22
	5.1 IMPLEMENTATION RESULTS	22
	5.1.1 SAMPLE QUERIES AND GENERATED RESPONSES	22
	5.1.2 TERMINAL EXECUTION OF THE RAG-BASED ASSISTANT	25
	5.2 DISCUSSION	30
6.	CONCLUSION AND FUTURE WORK	32
	6.1 CONCLUSION	32
	6.2 FUTURE WORK	32
	REFERENCES	34

LIST OF TABLES

TABLE NO.	TABLE NAME	PAGE NO.
5.1	Sample student queries and the generated RAG responses	23

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1	Architecture diagram of the system	13
4.1	Core workflow of the Enterprise Knowledge Assistant	16
4.2	Data ingestion and indexing pipeline	17
4.3	Query processing sequence of the assistant	18
4.4	Hybrid retrieval and answer grounding flow	19
5.1	Terminal execution for the GPA calculation query	26
5.2	Terminal execution for identifying active Industry MoUs of the CSE department	27
5.3	Terminal execution for the UG passing marks guidelines query	28
5.4	Terminal execution for the minimum attendance requirement query	29
5.5	Terminal execution for the laboratories and research facilities query	30

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CSE	Computer Science and Engineering
GPA	Grade Point Average
HOD	Head of the Department
LLM	Large Language Model
MoU	Memorandum of Understanding
NLP	Natural Language Processing
PDF	Portable Document Format
RAG	Retrieval-Augmented Generation
UG	Undergraduate

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Finding administrative and faculty details in a large educational institution can be a frustrating and time-consuming exercise. Students, particularly new ones, are forced to spend long hours searching the campus to find information about office addresses, room numbers or the contact details of faculty. This information is stored across physical notice boards, embedded in the website or distributed as PDF circulars, which makes the search task even more challenging. The burden of understanding the administrative structure of the institution falls upon the shoulders of new students and is exacerbated by the lack of centralized search resources. Although the portals are digital, keyword searching often fails to return direct results and ends up giving long lists of documents, links or PDFs that the user has to download and sift through to find a particular answer.

The problem arises from the need for an enterprise-level knowledge repository that can understand natural language queries and provide immediate, targeted answers. Recent advances in Artificial Intelligence (AI) and Natural Language Processing (NLP) have enabled conversational agents to act as information retrieval tools. However, standalone Large Language Models (LLMs) have the disadvantage of hallucinating, that is, generating answers based on their training data even when the information does not exist in the database.

Retrieval-Augmented Generation (RAG) overcomes this by using a pipeline that connects a document database containing verified institutional details with

an LLM, so that the generated answers are grounded in the database. This report presents an AI-Based Enterprise Knowledge Assistant for institutional information retrieval, which reduces the search time for both students and administrative staff by eliminating the need to sift through PDFs or search the campus.

1.2 OBJECTIVE

The main objective of the proposed system is to build an AI-Based Enterprise Knowledge Assistant that consolidates the scattered administrative and faculty information of an educational institution into a single, centralized knowledge base. The information handled by the system includes department locations, room numbers, faculty office addresses and contact details, which are otherwise distributed across physical notice boards, website pages and PDF circulars. The system allows users to search for these details in natural everyday language and to receive accurate and targeted answers instead of long lists of documents and links.

The system aims to ground every answer in verified institutional data. For this purpose, the institutional data is structured as high-dimensional vector embeddings and stored in a vector database, and Retrieval-Augmented Generation and semantic search are used to retrieve the most relevant context for a question. The objective also includes reducing the search time of students, especially new students, and reducing the repetitive queries that are directed to the administrative staff, so that the daily campus operations are improved and the overall student experience is enhanced.

1.3 EXISTING SYSTEM

In most educational institutions, administrative and faculty information is accessed through physical notice boards, pages on the institutional website and

PDF circulars. Students have to move around the campus or open multiple documents to find the office address, room number or contact details of a particular faculty member or department. New students are affected the most, since they are not yet familiar with the administrative structure of the institution, and the lack of a centralized search resource makes the process even more difficult.

Although the institutional portals are digital, they mainly depend on keyword searching. Keyword search often fails to return a direct result, and instead provides long lists of documents, links or PDFs, which the user has to download and read to find a single answer. Conversational agents based on standalone Large Language Models can understand natural language questions, but they generate answers from their training data and may therefore produce responses about information that does not exist in the institutional database, which is termed as hallucination.

Several RAG-based chatbots have also been proposed for education and university information, as reviewed in Chapter 2. These systems demonstrate that retrieval can reduce unsupported responses. However, most of them provide general university information or support teaching and policy queries, and there is still scope for improving the retrieval of highly specific details such as faculty contacts, room numbers, office locations and departmental details. Hence, a focused enterprise knowledge assistant for campus administrative and faculty information is highly demanded.

1.4 PROPOSED SYSTEM

The proposed system is an AI-Based Enterprise Knowledge Assistant that uses a Retrieval-Augmented Generation (RAG) architecture with a dual database design to answer natural language questions regarding campus faculty and administrative departments. The institutional administrative details and faculty

registers are consolidated into a centralized knowledge base, and the key textual attributes are structured as high-dimensional vector embeddings for efficient semantic retrieval.

As seen in the Figure 3.1, the validated dataset is stored in a relational Master Database for precise attribute lookup, in a Document Store, and, in the form of vector embeddings, in a Vector Database for semantic search. To preserve structural context such as faculty-department affiliations and physical campus layouts, the entity relationships are mapped into an interconnected Knowledge Graph that supports relational queries.

When a user asks a question in natural language through the web or mobile application, an embedding of the question is generated and used to perform semantic search and to identify the most relevant context from the database. The retrieved context is passed through the RAG pipeline, which synthesizes a direct answer with the relevant room or contact details. By eliminating the need to sift through PDFs or search the campus, the proposed assistant reduces the search time for both students and administrative staff. The rest of this report is organized as follows: Chapter 2 presents the literature survey on RAG-based chatbots, Chapter 3 describes the system design, Chapter 4 details the project description and the modules, Chapter 5 discusses the implementation and the results, and Chapter 6 concludes with the future work.

CHAPTER 2

LITERATURE SURVEY

This section of the report provides an all-round review about multiple existing systems that hold similar objectives and problem statements as that of the proposed project. The study of these existing papers will produce valuable insights for the construction of the proposed system.

2.1 LITERATURE REVIEW

Retrieval-Augmented Generation (RAG) has become a popular approach for developing knowledge-intensive conversational systems by combining information retrieval with generative language models. In (Lewis et al. 2020), the authors introduced the RAG architecture, where the relevant information is retrieved from an external knowledge source before generating the final response. This approach helps language models use additional information while answering questions and reduces their dependence on only the knowledge stored within the model.

The paper (Swacha and Gracel 2025) presented a survey on the use of RAG-based chatbots in education. The study discussed how RAG can be used to provide students with information from educational resources and institutional documents, and the use of retrieved information was found to be useful for reducing unsupported or hallucinated responses. However, the study mainly covers the broader use of RAG in education and does not specifically focus on retrieving detailed campus administrative information.

In (Lee 2025), the author proposed a RAG-based intelligent college information question-and-answer system. The system was designed to provide users with

college-related information through natural language questions. The study demonstrated that RAG can be used to connect college information sources with a conversational interface and provide more relevant answers. However, there is still scope for improving the retrieval of highly specific information such as faculty contacts, room numbers, office locations and departmental details.

The paper (N. P. et al. 2025), published in IJRASET, developed a RAG-based AI chatbot for student and institutional assistance using technologies such as semantic embeddings and local language models. The system aimed to reduce the effort required by students to search for common institutional information manually. The study showed that RAG can be applied to automate regular student queries using institutional documents. However, a more focused approach is required for retrieving specific administrative and location-based information from different campus resources.

The (Neupane et al. 2024) paper developed an informed chatbot for university resources to help students obtain information through natural language queries. The system focused on retrieving relevant university information and providing answers without requiring students to manually search through different resources. The study showed that retrieval-based approaches can improve access to university information. However, the system mainly focused on general university resources, leaving scope for more detailed retrieval of specific administrative information such as room numbers, faculty offices and department locations.

In (Olawore et al. 2025), the authors proposed a university chatbot using deep learning and a Retrieval-Augmented Generation approach. The study explored how RAG can be used to provide students with relevant answers from university information sources while reducing the possibility of unsupported responses. The work demonstrated the usefulness of combining language models with retrieved institutional information. However, the focus was mainly on chatbot-

based student support, with less emphasis on structured relationships between departments, faculty members, rooms and administrative offices.

The paper (Nguyen et al. 2025) investigated the integration of information retrieval techniques with Large Language Models for developing a document retrieval chatbot in an educational environment. Their work showed that retrieving relevant documents before generating an answer can improve the reliability of responses, and the approach is useful for handling information stored across multiple educational documents. However, document-based retrieval alone may not be sufficient for queries that require exact relationships between different entities, such as identifying the department of a faculty member or locating a particular office on campus.

The (Alwi Yafie et al. 2026) paper developed an academic chatbot using Retrieval-Augmented Generation for providing campus information services. The study demonstrated how RAG can be used to retrieve academic and institutional information and generate responses in a conversational manner, which can reduce the need for students to search through multiple sources manually. However, the system mainly addresses general campus information, while more focused retrieval of administrative details and location-specific information can still be improved.

In (Harsha Vardhan et al. 2026), the authors proposed AskUni, a RAG-based framework designed for intelligent university information systems. The system uses retrieval and generation techniques to provide users with relevant university information through natural language interaction. The study highlights the potential of RAG for making university information easier to access and reducing dependence on manual searches. However, the framework provides a general university information solution and does not specifically concentrate on connecting faculty, departments, rooms and administrative entities for more precise institutional queries.

The paper (Wu and Tseng 2025) developed a local generative AI teaching assistant using Retrieval-Augmented Generation technology to support the campus learning environment. The study focused on using locally processed institutional information with a generative AI system, which can help provide responses based on specific educational resources rather than relying only on general model knowledge. However, its primary focus is on teaching and learning support rather than administrative and campus-navigation queries such as faculty contacts, office locations and department details.

In (Peyton et al. 2025), the authors reviewed the use of university chatbots for student support and discussed how these systems have developed beyond simple frequently asked questions. The review highlights the use of chatbots for providing students with faster access to university services and information, and shows that conversational systems can reduce the effort required to find common institutional information. However, traditional chatbot systems may face difficulties when users ask detailed or less predictable questions, creating a need for systems that can retrieve information from updated institutional documents.

The systematic survey (Systematic Survey 2025) examined the use of Retrieval-Augmented Generation in educational applications. The study discussed different applications of RAG and highlighted its ability to combine information retrieval with generative models, where the responses are generated using information retrieved from external knowledge sources instead of depending only on the knowledge stored in the language model. The study also highlights the importance of retrieval quality and reliable knowledge sources for generating useful responses, which supports the need for a properly organized institutional knowledge base in campus information systems.

The paper (Hayrapetyan 2026) investigated the evaluation of Retrieval-Augmented Generation in a university policy chatbot. The study focused on

evaluating RAG responses without depending completely on manually labelled datasets, which is useful in university environments where policy documents contain large amounts of information and may be updated over time. The work highlights the importance of checking whether the generated responses are supported by the retrieved university information. However, the study mainly focuses on university policy information, whereas the proposed system also handles practical institutional details such as faculty contacts, department locations and room information.

In (Kavithra et al. 2025), the authors proposed a customized RAG support system for educational institutes. The study demonstrates how RAG can be adapted to the specific information requirements of an educational organization instead of using a general-purpose chatbot, and by using institutional information as the knowledge source, the system can provide more relevant responses to users. However, there is still scope for improving the organization and retrieval of highly specific campus information by combining semantic search with structured databases and relationships between different institutional entities.

The (Khan et al. 2025) paper developed an educational virtual assistant using Large Language Models and a Retrieval-Augmented Generation framework to provide information support to university students. The system retrieves relevant information from university sources before generating responses to queries related to areas such as admission processes, course selection and campus facilities. The study demonstrated that combining university knowledge sources with RAG can provide more relevant and updated responses to student queries. However, the system mainly focuses on general university assistance, while more specific institutional queries involving faculty contacts, department rooms, office locations and relationships between campus entities can be further addressed.

2.2 COMPARISON AND DISCUSSION

In the Section 2.1, it can be observed that the majority of the existing papers have prescribed the use of Retrieval-Augmented Generation for building conversational systems in educational environments (Lewis et al. 2020; Swacha and Gracel 2025). The studies have shown that retrieving relevant information from an external knowledge source before generating the response reduces the dependence of the model on its own stored knowledge and lowers the possibility of unsupported or hallucinated responses. The choice of the proposed system's core components, namely the institutional knowledge base, the embedding-based semantic retrieval and the RAG-based answer generation, is inspired by these papers. The studies also emphasise the importance of retrieval quality and reliable knowledge sources, which supports the need for a properly organized institutional knowledge base.

At the same time, most of the existing systems provide general university information, teaching and learning support or policy-related answers, and the review shows that document-based retrieval alone may not be sufficient for queries that need exact relationships between entities, such as identifying the department of a faculty member or locating a particular office on campus. Traditional chatbots also find it difficult to answer detailed or less predictable questions. Hence, the proposed system combines semantic search over vector embeddings with a relational Master Database and a Knowledge Graph of faculty, departments, rooms and administrative entities, so that specific details such as faculty contacts, room numbers and office locations can be retrieved accurately and the answers can be grounded in verified institutional data.

2.3 CONCLUSION

Though multiple RAG-based chatbots have the ability to answer student queries from university sources, a focused system that retrieves specific administrative

and location-based information, such as faculty contacts, room numbers, office locations and department details, is not widely available. Most of the existing systems concentrate on general university information, teaching support or policy documents, and they give less emphasis to the structured relationships between departments, faculty members, rooms and administrative offices. Hence, a system that consolidates the scattered institutional records into a single knowledge base, retrieves the exact details using semantic search and structured relationships, and generates answers that are grounded in verified institutional data, is needed.

CHAPTER 3

SYSTEM DESIGN

This chapter of the report briefly discusses the design of the proposed system with the help of the architecture diagram. The data ingestion and indexing process, and the semantic retrieval and response pipeline of the system are also discussed in detail.

3.1 SYSTEM ARCHITECTURE

The proposed system utilizes a Retrieval-Augmented Generation (RAG) architecture with a dual database design to answer natural language questions regarding campus faculty and administrative departments. The system acts as a centralized and intelligent gateway for navigating campus administrative and faculty records, and it unites Natural Language Processing, vector embeddings, semantic search and Retrieval-Augmented Generation to allow students to get instant answers using everyday language. The architecture of the entire system is provided in the Figure 3.1.

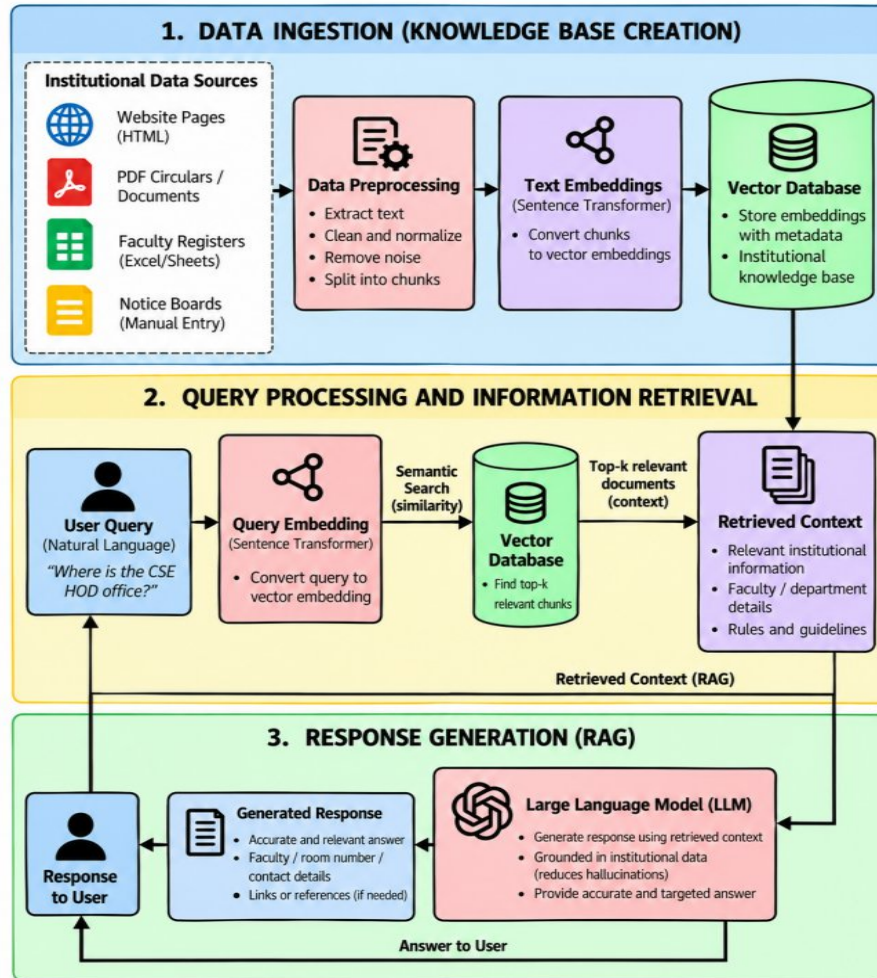


Figure 3.1. Architecture Diagram of the System

Figure 3.1. Architecture diagram of the system

The users interact with the system through a web or mobile application, which communicates with the API layer. The institutional data is maintained in a relational Master Database, a Document Store, a Vector Database and a Knowledge Graph, and the RAG pipeline uses the information retrieved from these stores to synthesize the final answer. This streamlines campus navigation, eliminating the hassle of digging through fragmented sources or constantly interrupting the staff for routine details.

3.2 DATA INGESTION AND INDEXING

The system is built using administrative details, such as departmental addresses and room numbers, and faculty contact details structured as entity tuples. The overall architecture is shown in Figure 3.1. The validated dataset is stored in a relational Master Database to allow precise attribute lookup, as well as in a Document Store.

Key textual attributes are also transformed into high-dimensional vector embeddings and stored in a Vector Database to allow semantic search. To preserve structural context, such as faculty-department affiliations and physical campus layouts, the entity relationships are mapped to an interconnected Knowledge Graph to aid in relational queries.

3.3 SEMANTIC RETRIEVAL AND RESPONSE PIPELINE

When a user asks a question through the web or mobile application, the API layer encodes the query into a vector. The system performs a hybrid search of semantic similarities from the Vector Database and a graph traversal through the Knowledge Graph to extract the exact relational details.

The extracted context is sent through the RAG pipeline, which synthesizes a direct answer with the relevant room or contact details. Since the answer is generated only from the retrieved institutional context, the response remains grounded in the verified data of the institution.

CHAPTER 4

PROJECT DESCRIPTION

4.1 METHODOLOGIES

This section provides a detailed view about the modules involved in the system, the processes within and the interaction between the modules. The system works on the administrative details and faculty registers of the institution, such as department directories, room locations, phone extensions, faculty rosters and office hours, which are consolidated into a single knowledge base. The system follows the core workflow of Ingest, Index, Retrieve and Generate. The system's architecture is thoroughly investigated in the below sub sections.

The structure of the system is subdivided into modules, where each module contains the processes that are linked or dependent on one another. The modules communicate with each other through the stored institutional data, the vector database and the RAG pipeline.

4.1.1 OVERALL WORKFLOW

The methodology of the proposed system is organised as a four-stage workflow of Ingest, Index, Retrieve and Generate, as shown in Figure 4.1. The first two stages form the offline phase, in which the institutional records are collected, cleaned and converted into searchable representations. The last two stages form the online phase, which is executed every time a student submits a question. Separating the two phases ensures that the computationally expensive preparation of the knowledge base is carried out only once, while the response to a query remains fast.

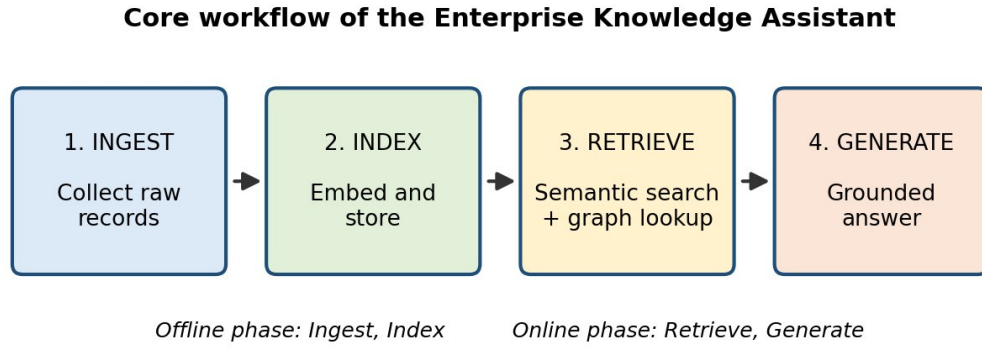


Figure 4.1. Core workflow of the Enterprise Knowledge Assistant

In the Ingest stage, the department directories, room locations, phone extensions, faculty rosters and office hours are gathered from notice boards, website pages and PDF circulars. In the Index stage, the validated records are stored in the relational Master Database and the Document Store, and the key textual attributes are converted into vector embeddings and stored in the Vector Database, while the entity relationships are mapped into the Knowledge Graph. In the Retrieve stage, the question of the student is embedded and matched against the stored representations. In the Generate stage, the retrieved context is passed to the language model, which produces a direct answer restricted to that context.

4.1.2 DATA INGESTION AND INDEXING PIPELINE

The data ingestion and indexing methodology is illustrated in Figure 4.2. Raw information is first collected from the three types of sources in which it is normally scattered, namely physical notice boards, institutional website pages, and PDF circulars or faculty registers. The collected records are consolidated into a single working set, so that the same fact is no longer repeated in several disconnected places.

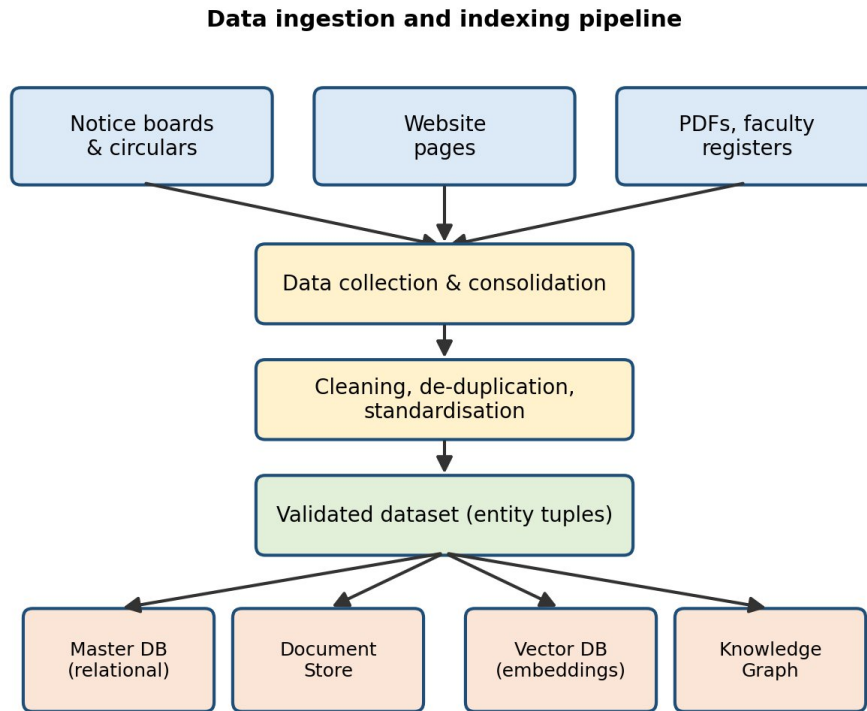


Figure 4.2. Data ingestion and indexing pipeline

The consolidated records then pass through a cleaning step. Outdated entries are removed, duplicated entries are merged and the formats of room numbers, phone extensions and names are standardised. The output of this step is a validated dataset of entity tuples, for example a faculty member, the department of the faculty member, the room, and the contact details. Since the vector retrieval and the generative components rely entirely on the accuracy of this dataset, the cleaning step directly affects the quality of every answer produced later.

The validated dataset is finally written to four stores, each serving a different purpose. The Master Database supports precise attribute lookup, the Document Store keeps the original text of the records, the Vector Database holds the embeddings used for semantic search, and the Knowledge Graph preserves relationships such as faculty-department affiliations and the physical layout of the campus. Using more than one store allows the system to answer both descriptive questions and exact relational questions.

4.1.3 QUERY PROCESSING SEQUENCE

The interaction between the components during the processing of a single question is shown as a sequence in Figure 4.3. The student types a question in everyday language into the web or mobile application, which forwards the query to the API layer. The API layer encodes the query into a vector using the same embedding approach that was used during indexing, so that the question and the stored records lie in the same vector space and can be compared directly.

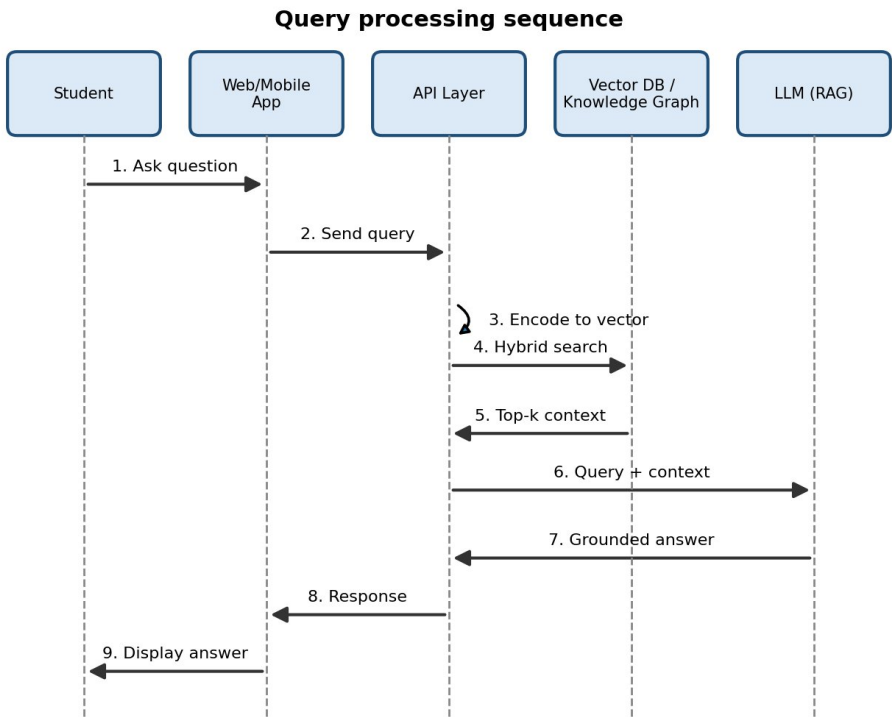


Figure 4.3. Query processing sequence of the assistant

The encoded query is used to perform a hybrid search over the Vector Database and the Knowledge Graph, which returns the most relevant context. The API layer then sends the original question together with this context to the language model of the RAG pipeline, receives the grounded answer and returns it to the application, where it is displayed to the student. Because the application, the API layer, the stores and the language model communicate only through well-

defined requests and responses, each component can be modified or replaced without affecting the others.

4.1.4 HYBRID RETRIEVAL AND ANSWER GROUNDING

Retrieval in the proposed system is hybrid in nature, as depicted in Figure 4.4. Semantic search over the Vector Database finds records whose meaning is close to the query even when the exact words differ, for example a question about the place to collect a hall ticket matching a record about the Examination Cell. Graph traversal over the Knowledge Graph, on the other hand, follows explicit relationships and is therefore suitable for questions that depend on exact links between entities, such as the department and the room of a particular faculty member.

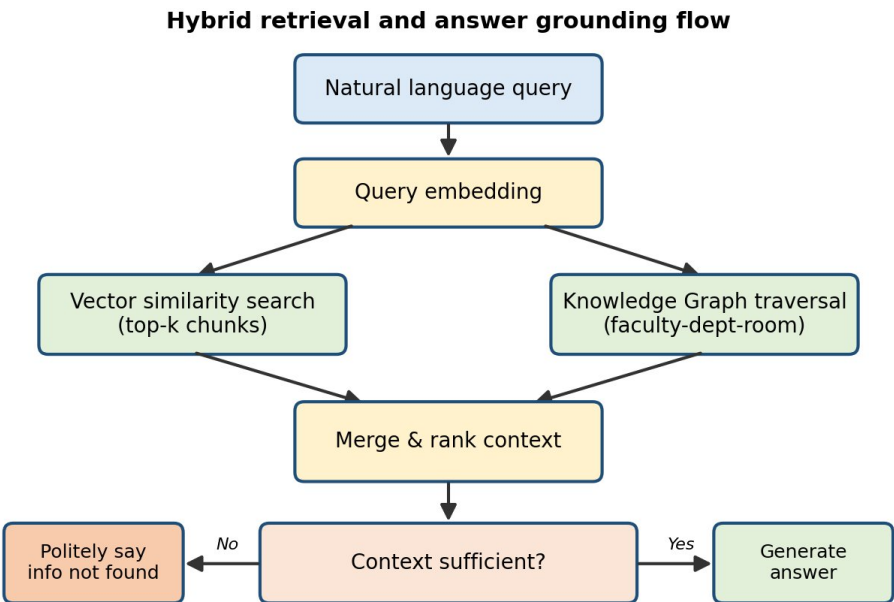


Figure 4.4. Hybrid retrieval and answer grounding flow

The results of the two searches are merged and ranked to form the final context. A sufficiency check is then applied on this context. If the context contains the

required information, it is passed to the language model, which is instructed to answer only from the supplied records. If the context does not contain the required information, the assistant replies that the information is not available instead of generating an unsupported answer. This restriction is the main safeguard against hallucination, and it ensures that every response given to the user can be traced back to verified institutional data.

4.2 MODULES

The system includes two major modules which in turn consist of several processes. These modules are isolated so as to support hassle free maintenance and updatation, and hence each module can be developed and tested independently.

- 1) Data Ingestion Module
- 2) RAG Response Generation Module

4.2.1 DATA INGESTION MODULE

The Data Ingestion Module functions as the backbone of the Enterprise Knowledge Assistant, gathering and organizing the scattered campus information. The main function of this module is to consolidate the raw institutional data, such as department directories, room locations, phone extensions, faculty rosters and office hours, into a single knowledge base.

To ensure accuracy in the system, the raw records are cleaned and standardized before being converted into a database. Outdated and duplicated records are filtered to maintain the integrity of the system, as the vector retrieval and the generative AI components rely on the accuracy of the dataset.

By collating the disconnected administrative files and notice board updates, this module eliminates the hassle of manual campus searches. The cleaned data is

formatted for vector embedding and semantic indexing to prepare it for instant retrieval. In this way, the module connects the raw institutional records to the vector database and the RAG system, allowing students to receive accurate answers to their everyday queries.

4.2.2 RAG RESPONSE GENERATION MODULE

The RAG Response Generation Module transforms the retrieved institutional data into clear and direct answers. Following the isolation of the relevant data from the vector database through semantic search, this module packages the details and the original natural language query of the student into context for the language model.

By restricting the context to the extracted information, the module synthesizes direct and accurate answers rather than generic summaries. For instance, if a student asks “Where is the Examination Cell?”, the module extracts the specific room record and generates an answer such as “Examination Cell is located in Room 204”.

This contextual grounding eliminates the hassle of downloading and reading through multi-page policy PDFs just to find a single address. Acting as the bridge between semantic retrieval and the user interface, the response generation module delivers instant responses to the students.

CHAPTER 5

IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULTS

The implementation of the proposed system along with the outputs is provided in this chapter. The outputs were obtained by testing the assistant with natural language queries of the type asked by students, and by executing the terminal-based RAG assistant over the institutional knowledge base. The following Table 5.1 presents the sample queries and the responses generated by the assistant.

5.1.1 SAMPLE QUERIES AND GENERATED RESPONSES

The Table 5.1 lists nine categories of queries, namely faculty location, administrative support, contact retrieval, service inquiries, exam cell services, library and resources, laboratory location, hostel administration and minimum credit requirement details. For each category, the input is a student query in natural language, and the output is the direct response generated by the RAG pipeline from the retrieved institutional data. It can be observed that the assistant returns the required room, office, contact or regulation details directly, instead of a list of documents or links.

Table 5.1. Sample student queries and the generated RAG responses

Query Category	Input (Student Natural-Language Query)	Output (Generated RAG Response)
Faculty Location	Where is Dr. Rajesh	Dr. Rajesh K. Sharma's office is

Query Category	Input (Student Natural-Language Query)	Output (Generated RAG Response)
	K. Sharma's cabin located and what is his email address?	located in Admin Block, 3rd Floor. His office hours are Monday – Friday from 10:00 AM – 12:00 PM. His Email Address is Rajesh.sharma@rajalakshmi.edu.in .
Administrative Support	Which office handles official transcript requests and where are they?	Transcript requests are processed by the Academic Affairs Office, located in the Main Administration Building, Room 102.
Contact Retrieval	How can I contact the Head of the Electrical Engineering Department?	You can reach the HOD of Electrical Engineering via email at ee.hod@institution.edu or call extension 4102 (Room 215, EE Block).
Service Inquiries	Where do I submit my fee waiver form?	Fee waiver forms must be submitted to the Student Accounts Desk in Room 108 of the Administrative Wing.
Exam Cell Services	Where do I apply for a duplicate hall ticket?	Duplicate hall tickets are issued at the Controller of Examinations Counter in the Exam Cell, Ground Floor, Central Block.
Library &	Who is in charge of	Digital library access is managed

Query Category	Input (Student Natural-Language Query)	Output (Generated RAG Response)
Resources	the digital library access and what is their email?	by Mr. Ramesh Kumar (Assistant Librarian). You can contact him at ramesh.lib@institution.edu.
Laboratory Location	Where is the Cyber Security and Digital Forensics Laboratory located and what tools does it have?	The laboratory is housed within the Department of Computer Science and Engineering. The laboratory is equipped with isolated network environments used for vulnerability analysis, penetration testing and digital forensics.
Hostel Administration	How can I reach the Chief Hostel Warden for room allotment issues?	The Chief Warden's office is at Block A Hostel Office, Room 002. Email: warden@institution.edu Phone: Ext. 5010.
Minimum Credit Requirement Details	What are the total minimum credit requirements to earn a B.E./B.Tech degree under Regulations 2023?	Under the regulations 2023, the credit requirements to earn a B.E./B.Tech degree is 160 to 165 credits.

5.1.2 TERMINAL EXECUTION OF THE RAG-BASED ASSISTANT

The proposed RAG-based assistant was also executed from the terminal for queries based on the institutional regulations and records. As seen in the

execution logs, the institutional knowledge base is indexed as 1,558 chunks in a vector database, the query is encoded using a sentence transformer model, the top three matching chunks are retrieved, and a local RAG synthesis engine running on the CPU generates the answer along with the source citations. The following Figures 5.1 to 5.5 show the execution of the assistant for five different queries.

The Figure 5.1 shows the execution of the terminal assistant for the query “How is the GPA calculated under the 2023 Autonomous Regulations?”. The system retrieves the relevant information from the institutional regulations and generates a source-grounded response explaining the GPA calculation method.

```
PS C:\Users\mithy> cd "C:\Users\mithy\OneDrive\Documents\AI BASED ENTERPRISE"
PS C:\Users\mithy\OneDrive\Documents\AI BASED ENTERPRISE> python terminal_assistant.py --query "How is the GPA calculated under the 2023 Autonomous Regulations?"
=====
RAJALAKSHMI ENGINEERING COLLEGE (REC) - TERMINAL ASSISTANT
Vector DB: ./chroma_db_storage | Index: 1,558 Chunks
=====
[System Check] Running on CPU (Check PyTorch CUDA Installation)
=====
[2026-09-08 11:18:28.792] [DEBUG] Starting initialization phase...
[2026-09-08 11:18:28.792] [DEBUG] Connecting to ChromaDB storage at 'C:\Users\mithy\OneDrive\Documents\AI BASED ENTERPRISE\chroma_db_storage'...
[2026-09-08 11:18:28.832] [DEBUG] Initializing SentenceTransformer 'all-MiniLM-L6-v2' on device 'cpu'...
Warning: You are sending unauthenticated requests to the Hf Hub. Please set a hf token to enable higher rate limits and faster downloads.
Loading weights: 100% | [DEBUG] Collection 'rec_institutional_data' ready with 1,558 indexed chunks (load time: 185.941s)
[2026-09-08 11:18:14.736] [DEBUG] Initialization finished in 185.941s.
[2026-09-08 11:18:14.736] [DEBUG] Received CLI query: "How is the GPA calculated under the 2023 Autonomous Regulations?"
[2026-09-08 11:18:14.736] [DEBUG] Encoding query and executing similarity search (Top-3) on CPU...
[2026-09-08 11:18:14.923] [DEBUG] Retrieved 3 chunks in 0.186s (Top similarity: 0.5685)
[2026-09-08 11:18:14.923] [DEBUG] Synthesizing answer from retrieved institutional context...
[2026-09-08 11:18:14.923] [DEBUG] Executing Local RAG Synthesis Engine on CPU...
[2026-09-08 11:18:14.924] [DEBUG] Local synthesis completed in 0.001s
[2026-09-08 11:18:14.924] [DEBUG] Total end-to-end query time: 0.188s
=====
QUESTION: How is the GPA calculated under the 2023 Autonomous Regulations?
=====
[GENERATED ANSWER] (Engine: Local RAG Synthesizer (CPU Engine))
Based on the official institutional records of Rajalakshmi Engineering College:
• GPA will be calculated in a similar manner, considering all the courses enrolled from
• GPA and during all the semesters in the case of CPA.
• The Grade Point Average (GPA) for the semester.
• The Cumulative Grade Point Average (CGPA) of all courses
• and to compute the Grade-Point Average (GPA). GPA is the ratio of the sum of
• 15.6. The credits earned through online courses with less than 3 credits shall not be considered
=====
SOURCE CITATIONS (Top 3 Matching Chunks):
=====
[1] Source: Academics/REC-2023-UG-Regulations.pdf | Page: 28
Section: REC-2023-UG-Regulations - GPA will be calculated in a similar manner, considering all the courses enrolled from
Similarity: 0.5685
Context: "28 GPA will be calculated in a similar manner, considering all the courses enrolled from first semester. "UG" and "SA" grades will be excluded for calculating GPA and CGPA. 15.6. The credits earned th..."
[2] Source: Academics/REC-2023-PG-Regulations.pdf | Page: 17
Section: REC-2023-PG-Regulations - where
Similarity: 0.5609
Context: "17 where + Ci - is the Credits assigned to the course + GPI - is the grade point corresponding to the letter grade obtained in that course + n - is number of Courses successfully cleared during the pa..."
[3] Source: Academics/REC-2023-UG-Regulations.pdf | Page: 19
Section: REC-2023-UG-Regulations - 15.2 For Mandatory Non-credit Courses, a Pass / O (Re-appearance) will appear in the
Similarity: 0.5679
Context: "contain the following details: + The Name of the college and affiliating university. + The list of courses registered during the semester and the grades scored. + The Grade Point Average (GPA) for the..."
=====
```

Figure 5.1. Terminal execution for the GPA calculation query

The Figure 5.2 shows the terminal execution of the proposed RAG-based assistant for identifying the active Industry MoUs of the Computer Science and Engineering department. The system retrieves the relevant institutional documents using similarity search and generates a source-grounded response along with document and page-level citations, demonstrating reliable and traceable information retrieval.

```

PS C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE> python terminal_assistant.py --query "Which companies have active Industry MoUs with the Computer Science and Engineering department?"
=====
RAJALAKSHMI ENGINEERING COLLEGE (REC) - TERMINAL ASSISTANT
Vector DB: ./chroma_db_storage | Index: 1,558 Chunks
=====
[System Check] Running on CPU (Check PyTorch CUDA installation)
=====
[2026-09-08 11:21:07.448] [DEBUG] Starting initialization phase...
[2026-09-08 11:21:07.451] [DEBUG] Connecting to ChromaDB storage at 'C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE\chroma_db_storage'...
[2026-09-08 11:21:07.610] [DEBUG] Initializing SentenceTransformer 'all-MiniLM-L6-v2' on device='cpu'...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 103/103 [00:00:00.00, 1218.12it/s]
[2026-09-08 11:21:24.830] [DEBUG] Collection 'rec_institutional_data' ready with 1,558 indexed chunks (Load time: 17.356s)
[2026-09-08 11:21:24.836] [DEBUG] Initialization finished in 17.356s.
[2026-09-08 11:21:24.837] [DEBUG] Received CLI query: "Which companies have active Industry MoUs with the Computer Science and Engineering department?"
[2026-09-08 11:21:24.837] [DEBUG] Encoding query and executing similarity search (Top-3) on CPU...
[2026-09-08 11:21:24.927] [DEBUG] Retrieved 3 chunks in 0.090s (Top similarity: 0.6063)
[2026-09-08 11:21:24.927] [DEBUG] Synthesizing answer from retrieved institutional context...
[2026-09-08 11:21:24.928] [DEBUG] Executing Local RAG Synthesis Engine on CPU...
[2026-09-08 11:21:24.928] [DEBUG] Local synthesis completed in 0.001s
[2026-09-08 11:21:24.928] [DEBUG] Total end-to-end query time: 0.092s
=====
QUESTION: Which companies have active Industry MoUs with the Computer Science and Engineering department?
=====
[GENERATED ANSWER] (Engine: Local RAG Synthesizer (CPU Engine))

Based on the official institutional records of Rajalakshmi Engineering College:

• The department has established active industry alliances with global tech leaders:
• The Industry-Institute Interaction is further strengthened time to time by initiating MoUs
• (Corporate entities) etc. and how they have contrib uted to the development of the
MoUs FOR STRENGTHENING INDUSTRY INSTITUTE INTERACTION
• MoUs with Sutherland Global
• Sutherland Global services; Regional Telecom Traini ng Centre, Chennai and Mitachi solutions

=====
SOURCE CITATIONS (Top 3 Matching Chunks):
=====
[1] Source: Administration/NAAC-558-2016.pdf | Page: 8
Section: NAAC-558-2016 - Rajalakshmi Engineering College | 8
Similarity: 0.6063
Context: "The Industry-Institute interaction is further strengthened time to time by initiating MoUs for a specified period with reputed organizations s uch as IBM Global Services; Chennai based Super Auto Forg..."

[2] Source: Administration/NAAC-558-2016.pdf | Page: 129
Section: NAAC-558-2016 - Rajalakshmi Engineering College | 129
Similarity: 0.6286
Context: "Institutions of national importance/other universit ies/Industries/Corporate (Corporate entities) etc. and how they have contrib uted to the development of the institution. MoUs FOR STRENGTHENING INDU..."

[3] Source: Departments/Computer_Science_and_Engineering.md
Section: 1. Industry Collaborations & Memorandums of Understanding (MoUs)
Similarity: 0.5931
Context: "The department has established active industry alliances with global tech leaders: - **IBM** Center of Excellence for Cloud and Big Data analytics. - **Infosys Campus Connect** Industry-aligned foun..."
=====
PS C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE>

```

Figure 5.2. Terminal execution for identifying active Industry MoUs of the CSE department

The Figure 5.3 shows the terminal execution of the proposed RAG-based assistant for retrieving the UG passing marks guidelines. The system searches the institutional knowledge base, retrieves the relevant academic regulations, and generates a source-grounded response describing the required passing marks and the assessment criteria.

```

PS C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE> python terminal_assistant.py --query "What are the UG passing marks guidelines?"
=====
RAJALAKSHMI ENGINEERING COLLEGE (REC) - TERMINAL ASSISTANT
Vector DB: ./chroma_db_storage | Index: 1,558 Chunks
=====
[System Check] Running on CPU (Check PyTorch CUDA installation)
=====
[2026-09-08 11:24:33.445] [DEBUG] Starting initialization phase...
[2026-09-08 11:24:33.446] [DEBUG] Connecting to ChromaDB storage at 'C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE\chroma_db_storage'...
[2026-09-08 11:24:33.561] [DEBUG] Initializing SentenceTransformer 'all-MiniLM-L6-v2' on device='cpu'...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 103/103 [00:00:00.00, 2485.36it/s]
[2026-09-08 11:24:54.908] [DEBUG] Collection 'rec_institutional_data' ready with 1,558 indexed chunks (Load time: 21.462s)
[2026-09-08 11:24:54.908] [DEBUG] Initialization finished in 21.463s.
[2026-09-08 11:24:54.908] [DEBUG] Received CLI query: "What are the UG passing marks guidelines?"
[2026-09-08 11:24:54.908] [DEBUG] Encoding query and executing similarity search (Top-3) on CPU...
[2026-09-08 11:24:55.002] [DEBUG] Retrieved 3 chunks in 0.090s (Top similarity: 0.6088)
[2026-09-08 11:24:55.003] [DEBUG] Synthesizing answer from retrieved institutional context...
[2026-09-08 11:24:55.003] [DEBUG] Executing Local RAG Synthesis Engine on CPU...
[2026-09-08 11:24:55.003] [DEBUG] Local synthesis completed in 0.001s
[2026-09-08 11:24:55.003] [DEBUG] Total end-to-end query time: 0.095s
=====
QUESTION: What are the UG passing marks guidelines?
=====
[GENERATED ANSWER] (Engine: Local RAG Synthesizer (CPU Engine))

Based on the official institutional records of Rajalakshmi Engineering College:

• 14.4. The passing requirement for the mandatory courses which are assessed only through
• Minimum Passing Standard:** A candidate must secure not less than **50% marks in End Semester Examinations** and a minimum **Grade Point of 6 (Letter Grade B)** overall in each course.
• onwards, the continuous assessment marks will not be considered and he/she has to secure
• Second Class:** All other candidates completing the program within 4 years.
• Continuous Internal Assessment (CIA):** **40 Marks**
• Continuous Assessment Tests (CAT 1 & CAT 2): 20 Marks

```

Figure 5.3. Terminal execution for the UG passing marks guidelines query

The Figure 5.4 shows the terminal execution of the proposed RAG-based assistant for retrieving the minimum attendance requirement for the end-semester examinations. The system searches the institutional knowledge base, retrieves the relevant regulatory information, and generates a source-grounded response based on the official college records.

```
PS C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE> python terminal_assistant.py --query "What is the minimum attendance percentage required to appear for end-semester examinations?"
=====
RAJALAKSHMI ENGINEERING COLLEGE (REC) - TERMINAL ASSISTANT
Vector DB: ./chroma_db_storage | Index: 1,558 Chunks
=====
[System Check] Running on CPU (Check PyTorch CUDA installation)
=====
[2026-09-08 11:29:38.847] [DEBUG] Starting initialization phase...
[2026-09-08 11:29:38.847] [DEBUG] Connecting to ChromaDB storage at 'C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE\chroma_db_storage'...
[2026-09-08 11:29:39.146] [DEBUG] Initializing SentenceTransformer 'all-MiniLM-L6-v2' on device='cpu'...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|#####| 183/183 [00:00<00:00, 2784.44it/s]
[2026-09-08 11:30:09.016] [DEBUG] Collection 'rec_institutional_data' ready with 1,558 indexed chunks (Load time: 38.168s)
[2026-09-08 11:30:09.016] [DEBUG] Initialization finished in 38.169s.
[2026-09-08 11:30:09.016] [DEBUG] Received CLI query: "What is the minimum attendance percentage required to appear for end-semester examinations?"
[2026-09-08 11:30:09.016] [DEBUG] Encoding query and executing similarity search (Top-3) on CPU...
[2026-09-08 11:30:09.192] [DEBUG] Retrieved 3 chunks in 0.176s (Top similarity: 0.7346)
[2026-09-08 11:30:09.192] [DEBUG] Synthesizing answer from retrieved institutional context...
[2026-09-08 11:30:09.192] [DEBUG] Executing Local RAG Synthesis Engine on CPU...
[2026-09-08 11:30:09.193] [DEBUG] Local synthesis completed in 0.001s
[2026-09-08 11:30:09.193] [DEBUG] Total end-to-end query time: 0.177s
=====
QUESTION: What is the minimum attendance percentage required to appear for end-semester examinations?
=====
[GENERATED ANSWER] (Engine: Local RAG Synthesizer (CPU Engine))

Based on the official institutional records of Rajalakshmi Engineering College:

• 7.4 For the purpose of calculation of attendance requirement for writing the end semester
• but having at least 50% of attendance in every course of that semester can appeal to the
• Therefore, he/she shall secure not less than 75% attendance (after rounding off to the
• examinations by the students for every course, the following method shall be used.
• 7.5. A student who has not satisfied the attendance requirement as per clauses from 7.1 to 7.4,
• 7. REQUIREMENTS FOR COMPLETION OF THE SEMESTER

=====
SOURCE CITATIONS (Top 3 Matching Chunks):
```

Figure 5.4. Terminal execution for the minimum attendance requirement query

The Figure 5.5 shows the terminal execution of the proposed RAG-based assistant for retrieving information about the major laboratories and research facilities. The system performs similarity-based retrieval from the institutional knowledge base and generates a response describing the available research centres, laboratories and research-related facilities.

```

PS C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE> python terminal_assistant.py --query "What are the major laboratories and research facilities available in the department?"
=====
RAJALAKSHMI ENGINEERING COLLEGE (REC) - TERMINAL ASSISTANT
Vector DB: ./chroma_db_storage | Index: 1,558 Chunks
=====
[System Check] Running on CPU (Check PyTorch CUDA installation)
=====
[2026-09-08 11:32:42.539] [DEBUG] Starting initialization phase...
[2026-09-08 11:32:42.539] [DEBUG] Connecting to ChromaDB storage at 'C:\Users\nithy\OneDrive\Documents\AI BASED ENTERPRISE\chroma_db_storage'...
[2026-09-08 11:32:42.755] [DEBUG] Initializing SentenceTransformer 'all-MiniLM-L6-v2' on device='cpu'...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 103/103 [00:00:00:00, 2804.77it/s]
[2026-09-08 11:33:15.868] [DEBUG] Collection 'rec_institutional_data' ready with 1,558 indexed chunks (Load time: 33.329s)
[2026-09-08 11:33:15.869] [DEBUG] Initialization finished in 33.330s.
[2026-09-08 11:33:15.869] [DEBUG] Received CLI query: "What are the major laboratories and research facilities available in the department?"
[2026-09-08 11:33:15.869] [DEBUG] Encoding query and executing similarity search (Top-3) on CPU...
[2026-09-08 11:33:15.977] [DEBUG] Retrieved 3 chunks in 0.108s (Top similarity: 0.6268)
[2026-09-08 11:33:15.978] [DEBUG] Synthesizing answer from retrieved institutional context...
[2026-09-08 11:33:15.978] [DEBUG] Executing Local RAG Synthesis Engine on CPU...
[2026-09-08 11:33:15.978] [DEBUG] Local synthesis completed in 0.001s
[2026-09-08 11:33:15.979] [DEBUG] Total end-to-end query time: 0.109s
=====
QUESTION: What are the major laboratories and research facilities available in the department?
=====
[GENERATED ANSWER] (Engine: Local RAG Synthesizer (CPU Engine))

Based on the official institutional records of Rajalakshmi Engineering College:

• Many faculty members are carrying out their research in core and applied research
• areas in these centers and full-time/part-time Ph.D. / M.S. (by Research) scholars are
• The Departments identified as Research Centres are well equipped with state-of-the
• The laboratory and library facilities available in the campus can be utilized by the staff
• facilities of the institution by its staff and students?
• and students involved in research with no time restriction. Log books are also available to

```

Figure 5.5. Terminal execution for the laboratories and research facilities query

5.2 DISCUSSION

The outputs that had been generated over the course of the implementation provided vital information regarding the quality of the retrieval and the methods used. The assistant was able to answer the queries on faculty locations, administrative offices, contact details, laboratories and academic regulations directly in natural language, and the terminal executions show that the answers are accompanied by the source documents and the page-level citations, which makes the responses traceable to the official institutional records. The retrieval of only the top matching chunks from the vector database, instead of searching through entire documents, made it possible to generate the responses within a fraction of a second after initialization, as seen in the execution logs.

The sample interactions of the assistant are listed below, which show how a student can obtain the required details in everyday language without searching the campus or sifting through fragmented sources.

1. **Input:** A student asks, “Who is the faculty coordinator for the CSE department” **Output:** The assistant provides the faculty coordinator’s name and the relevant contact details from the college records.
2. **Input:** A student asks, “Where is the Exam Cell located” **Output:** The assistant gives the correct building, room, or office location of the Exam Cell.
3. **Input:** A student asks, “Where can I find information about college regulations” **Output:** The assistant provides the relevant regulation details from the uploaded college documents.

All of these implementations add value to the overall system and work towards eliminating the need to constantly interrupt the administrative staff for routine details, while keeping the answers grounded in verified institutional data.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

In this report, we have discussed the problem statement and objectives of the proposed system, and have constructed a system that provides a solution to the problem of finding administrative and faculty details in an educational institution. The AI-Based Enterprise Knowledge Assistant provides a centralized, intelligent gateway for navigating the campus administrative and faculty records. By uniting Natural Language Processing, vector embeddings, semantic search and Retrieval-Augmented Generation, the platform allows students to get instant answers using everyday language. This streamlines campus navigation, eliminating the hassle of digging through fragmented sources or constantly interrupting the staff for routine details. The design of the proposed system, the modules, the implementation and the results have been discussed in detail in this report.

6.2 FUTURE WORK

The proposed system provides a centralized and instant knowledge repository for campus information. But the real-world usage of the proposed system poses further challenges. Since the institutional circulars, faculty registers and department details change over time, in the future, the knowledge base can be extended with mechanisms to keep the records updated and to filter outdated information automatically. The retrieval of highly specific information can also be further improved by combining semantic search with structured databases

and by extending the relationships between faculty, departments, rooms and administrative entities in the Knowledge Graph.

Apart from this, the future work also includes extending the knowledge base to cover more institutional documents, deploying the assistant as a web and mobile application for wider reach, and evaluating the system with a larger set of real student queries and user studies, to measure the reduction in the search time of students and in the repetitive queries to the administrative staff. Hence, making the system more reliable, more complete and more adaptable to newer models and techniques is the primary objective for the future scope of the system.

REFERENCES

1. P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020.
2. J. Swacha and M. Gracel, “Retrieval Augmented Generation (RAG) Chatbots for Education: A Survey of Applications,” *Applied Sciences*, vol. 15, no. 8, p. 4234, 2025. doi: 10.3390/app15084234.
3. H. Lee, “RAG-based Intelligent College Information Question and Answer System,” *Journal of Internet of Things and Convergence*, vol. 11, no. 3, pp. 161–166, 2025.
4. N. P., A. R. Arohan, A. P. C. Adhithyan, M. Suhail, S. B. Muhammed, and L. V. U., “RAG-Based AI Chatbot for Student and Institutional Assistance,” *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 13, no. 9, pp. 858–863, 2025. doi: 10.22214/ijraset.2025.73970.
5. S. Neupane, E. Hossain, J. Keith, H. Tripathi, F. Ghiasi, N. A. Golilarz, A. Amirlatifi, S. Mittal, and S. Rahimi, “From Questions to Insightful Answers: Building an Informed Chatbot for University Resources,” *2024 IEEE Frontiers in Education Conference (FIE)*, 2024, doi: 10.1109/FIE61694.2024.10892994.
6. K. Olawore, M. McTear, and Y. Bi, “Development and Evaluation of a University Chatbot using Deep Learning: A RAG-Based Approach,” *Conference Proceedings*, 2025, doi: 10.1007/978-3-031-88045.
7. D. D. K. Nguyen, V. K. Mach, T. L. D. Le, and C. Pham-Nguyen, “Integrating Information Retrieval and LLMs: A Document Retrieval Chatbot in Education Settings,” *Intelligent Systems and Data Science (ISDS 2025)*, pp. 461–475, 2025.
8. M. Alwi Yafie, A. U. Zailani, W. Muttaqin, M. S. A. Daffansyah, and M. R. Ramzi, “Academic Chatbot for Campus Information Services Using Retrieval-Augmented Generation,” *Sinkron: Jurnal dan Penelitian Teknik Informatika*, vol. 10, no. 3, 2026, doi: 10.33395/sinkron.v10i3.16088.
9. K. S. Harsha Vardhan, K. Pratham, C. Jeevan, G. Sekhar Reddy, and G. L. Anand Babu, “AskUni: A Retrieval-Augmented Generation Framework for Intelligent University Information Systems,” *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 13, no. 3, pp. i384–i389, 2026.
10. J.-W. Wu and M.-H. Tseng, “Developing a Local Generative AI Teaching Assistant System: Utilizing Retrieval-Augmented Generation Technology to Enhance the Campus Learning Environment,” *Electronics*, vol. 14, no. 17, p. 3402, 2025, doi: 10.3390/electronics14173402.

11. K. Peyton, S. Unnikrishnan, and B. Mulligan, “A Review of University Chatbots for Student Support: FAQs and Beyond,” *Discover Education*, vol. 4, article 21, 2025.
12. “Retrieval-Augmented Generation for Educational Application: A Systematic Survey,” *Computers and Education: Artificial Intelligence*, vol. 8, p. 100417, 2025, doi: 10.1016/j.caeai.2025.100417.
13. L. Hayrapetyan, “Label-Free Evaluation of Retrieval-Augmented Generation in a University Policy Chatbot,” *Proceedings of the 57th ACM Technical Symposium on Computer Science Education (SIGCSE TS 2026)*, vol. 2, pp. 1665–1666, 2026, doi: 10.1145/3770761.3777161.
14. K. S. Kavithra, A. T. Abirami, D. Dharun, K. Kishore, and M. V. S. Mohanraj, “Customized RAG Support System for Educational Institutes,” *Proceedings of the 2025 International Conference*, 2025.
15. U. H. Khan, M. H. Khan, and R. Ali, “Large Language Model based Educational Virtual Assistant using RAG Framework,” *Procedia Computer Science*, vol. 252, pp. 905–911, 2025, doi: 10.1016/j.procs.2025.01.051.